

Semantic Radar Grids

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Abstract—Radar has a long tradition in driver assistance systems. One of its advantages is its robustness against certain light and weather conditions. It is usually used to track surrounding objects and to detect objects which can be utilized for assistance functions, e.g. for emergency braking. However for autonomous driving, a deeper understanding of the vehicle's surroundings is necessary. In this paper we show that semantic knowledge can be obtained from a radar grid by classifying the containing objects on the cell level. This allows for omitting a prior object extraction and results directly in a semantic radar grid. We show on a data set recorded with four radar sensors that this approach produces good results distinguishing cars and other objects.

I. INTRODUCTION

In recent years, driver assistance systems have evolved, but the driver still has to constantly monitor the vehicle and intervene if the system fails. To get the driver out of the loop and enable autonomous driving, the car needs, among other things, a reliable understanding of the vehicle's surrounding. A semantic representation of the environment is important, especially in semi-structured environments [1]. The semantic environment perception can help landmark-based localization systems [2] to decide which landmarks are reliable and which not. Further, it enables path planing, e.g. by finding an appropriate parking spot or providing basis for understanding the scene.

Traditional object classification in radar had two main fields: The surveillance of airborne objects [3] or the detection of objects using airborne radar, for example for military reasons [4] or for automated mapping [5].

Recent improvements in automotive radar enabled object classification for autonomous driving. Due to its robustness with respect to weather conditions and its good spatial resolution, radar can provide redundancy and enhance results from camera classification.

Previous work, e.g. [6], focused on classifying pedestrians using the characteristic Doppler spectrum caused by the velocity differences between the legs and the torso of a walking person.

To classify the static world, Dube et al. [7] introduced a method for detecting parked vehicles in radar grids. Lombacher et al. [8] suggested a snippet based approach to make the classification rotation independent because the orientation of the objects in the maps is nearly arbitrary.

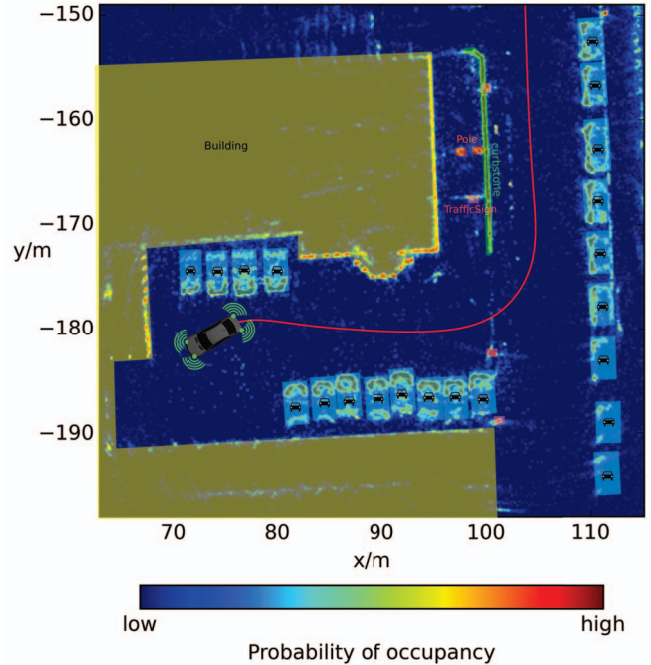


Fig. 1. Vehicle sensor configuration (green) inside an occupancy grid. The objects in this occupancy grid are labeled manually.

When the classification is done object-wise, the data first has to be clustered into objects, using a clustering algorithm, connected component, watershed, etc. There are many scenarios where such a clustering algorithm is error-prone. Sometimes, objects (e.g. vehicles) may be segmented into multiple clusters or multiple objects may be merged into a single cluster, which is therefore hard to classify. These clustering algorithms are highly dependent on parameters which are usually chosen through expert knowledge.

In image processing, more advanced object segmentation algorithms have been developed, e.g. based on bounding boxes [9] or pixel-wise [10], where a learning algorithm also determines the segmentation. Another approach is to classify the scene cell-wise as has been done for images [11], [12].

In this paper, we show that a similar approach can be used for radar grids. The next chapter introduces the data set including the preprocessing and the labeling. Section III explains the classification method. The results are shown in

Section IV.

II. DATA

In this paper, a custom data set is used, as there is currently no public radar data set available. The data set was recorded on different urban scenarios having the focus on parking spaces and side roads and distinguish between cars and non-cars.

For data recording, four radar sensors were installed on a test vehicle. The sensors were mounted at the four corners of the vehicle as depicted in Fig. 1, allowing for circumferential perception. The radar sensors operate at 76 GHz and have a coverage of up to 40 m with a range accuracy better than 0.15 m . The field of view of a single sensor is 140° with an angular accuracy better than 1° . The position of the vehicle is estimated by dead reckoning using the vehicle's odometry.

A. Occupancy Radar Grids

Radar data is received as measurements Z_t for a given time t where targets have already been extracted by the sensor. The state of each target is given by a radial distance r , an azimuth angle θ , a radial velocity v_r and a normalized amplitude a which is proportional to the radar cross section of the target. Further, the variance of all spatial and velocity information is given. The position and orientation X_t of the vehicle is determined by its internal odometry unit using the turning rate of the wheels and the yaw rate sensor.

For classification purposes, only stationary objects should be represented in the radar grid. To discard measurements from moving objects, the radial velocity of the measurement v_r is used. This is done by computing the expected radial velocity $v_{r,\text{static}}$ of static objects due to the ego-motion. Knowing the motion of the ego vehicle in the rotation center v_x^{ego} , v_y^{ego} including the yaw rate ω^{ego} the velocity of the sensor v_x^{sen} , v_y^{sen} can be computed knowing its mounting position s_x , s_y . For non-drifting vehicles the velocity v_y^{ego} is 0.

$$\begin{pmatrix} v_x^{\text{sen}} \\ v_y^{\text{sen}} \end{pmatrix} = \begin{pmatrix} \omega^{\text{ego}} s_y - v_{x_{0,0}}^{\text{ego}} \\ \omega^{\text{ego}} s_x \end{pmatrix} \quad (1)$$

With the sensors installation angle α and the angle of the measurement θ the expected radial and tangential velocity v_r and v_t of static objects can be computed by the rotation matrix.

$$\begin{pmatrix} v_{r,\text{static}} \\ v_{t,\text{static}} \end{pmatrix} = \begin{pmatrix} \cos(\alpha + \theta) & -\sin(\alpha + \theta) \\ \sin(\alpha + \theta) & \cos(\alpha + \theta) \end{pmatrix} \begin{pmatrix} v_x^{\text{sen}} \\ v_y^{\text{sen}} \end{pmatrix} \quad (2)$$

Only v_r is measured and thus all targets with a velocity higher than $3\sigma_{v_r}$ of the co-variance are excluded. Targets with only a tangential velocity cannot be filtered, however this is only true for a very limited number of dynamic targets.

$$Z^{\text{static}} = \{z \in Z \mid |v_r - v_{r,\text{static}}| < 3\sigma_{v_r}\} \quad (3)$$

Current automotive radar sensors provide merely a few targets per measurement cycle. It is therefore intractable to classify objects on such sparse data. To overcome this problem, the data is accumulated in occupancy grids as

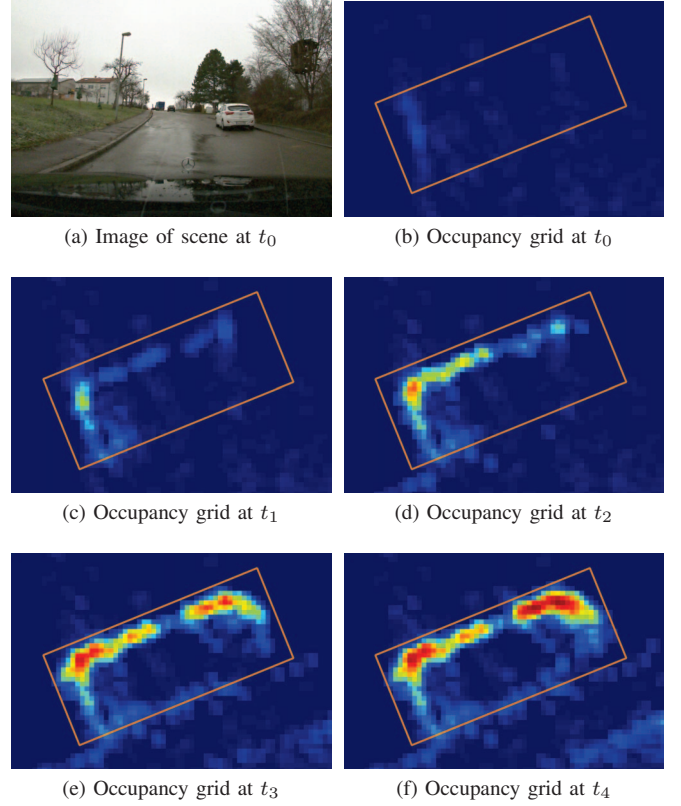


Fig. 2. The figure shows the labels on the gridmap over the time. The snippets (b)-(f) contain the label of the white car in (a). The orange label was placed manually at t_4 and transferred using the global coordinates.

introduced by Elfes [13] and suggested for radar by Werber et al. [14].

The occupancy grid, $M = \{m^{[1]}, m^{[2]}, \dots, m^{[N]}\}$, consists of N grid cells $m^{[n]}$. Each cell can be either occupied or free. The cells are assumed to be independent of each other and according to [15] each cell of the occupancy grid is associated with the time-dependent posterior probability that the cell is occupied at time t :

$$p(M_t | Z_{0:t}^{\text{static}}, X_{0:t}) = \prod_n p(m^{[n]} | Z_{0:t}^{\text{static}}, X_{0:t}) \quad (4)$$

These grids therefore contain the accumulated measurement information and are used as input for the classifier.

The cell size of the radar grid is $0.1\text{ m} \times 0.1\text{ m}$ and the overall size is $100\text{ m} \times 100\text{ m}$. This grid is moved so that the ego vehicle stays in the center. A snapshot of the map is stored every second.

B. Labels

The data is labeled manually. For this purpose, the labeling person annotates the area in the radar grid with the help of an additional surround view video stream. A variety of classes have been chosen for labeling, but due to the limited amount of data we consider only *car*, *other* and *not-labeled* for this paper.

As depicted in Fig. 2, the objects are labeled on the grid where the ego vehicle has already passed and all

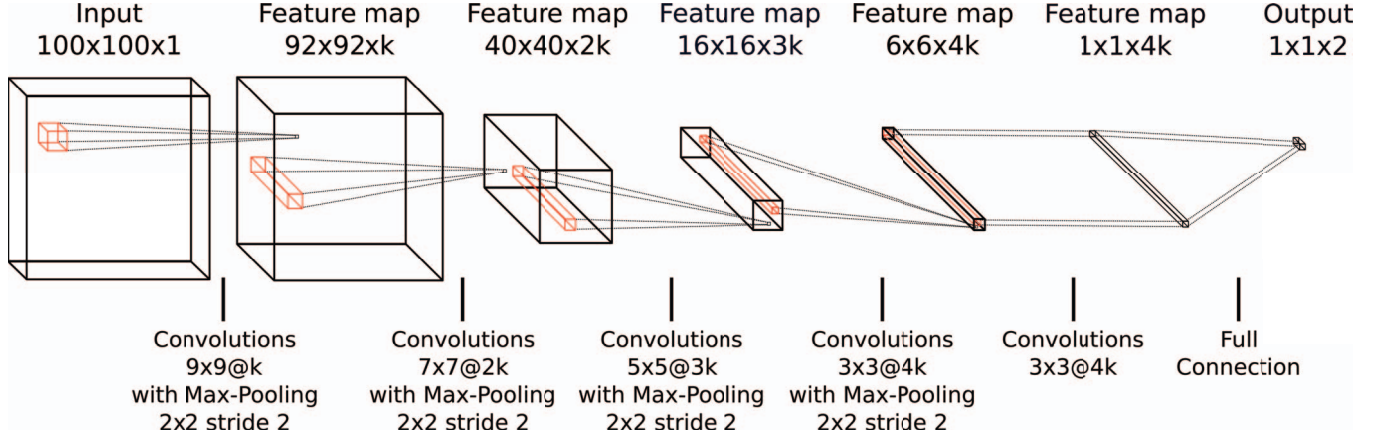
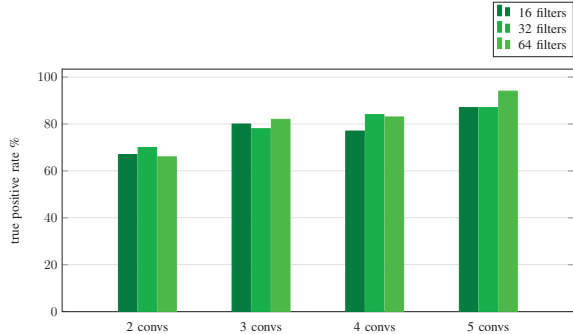
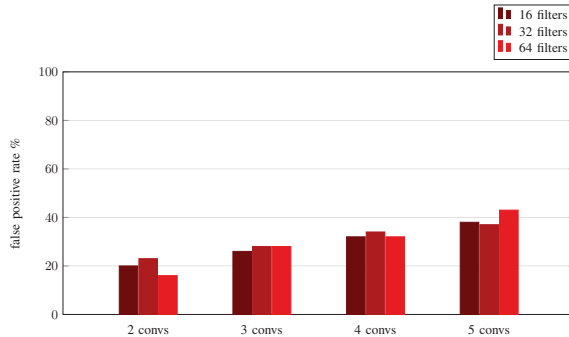


Fig. 3. Network structure of the largest applied CNN. The filter size is given by [filter width]x[filter height]@[no of filters]. For each convolutional layer the stride is 1.



(a) True positive rates



(b) False positive rates

Fig. 4. Pixel-wise true vs. false positive rates, regarding cars, for classifiers with two to five convolutional layers and with 16, 32 and 64 filters in the first layer. The car occupancy threshold is set to 0.15. The bars are grouped by the number of convolutional layers.

measurements were integrated in the grid in Fig. 2f. Those labels are created in a global coordinate systems, and can therefore be used while the measurements of an object are still integrated into the grid 2b-2e. Thus, some area of a radar grid contains labels of objects that are not yet fully visible or not visible at all $t < t_0$. This leads to empty or poorly built up samples. To resolve the issue on which observation level a cell should be classified, a threshold T_{occ} is introduced, so a label is only valid if the probability of the cell is greater

than the threshold ($T_{occ} < P(M^n)$) otherwise the label *not-labeled* is applied.

C. Data set

TABLE I
OCCURRENCES OF THE LABELED CLASSES IN THE DATA SET.

Class	Occurrences	Pixels
Car	262,858	(0.36 %)
Other	586,944	(0.80 %)
not-labeled	72,043,528	(98.84 %)

The training data is divided into training (60%), validation (20%) and test-set (20%) based on different recorded sequences. So the inputs from the different sets are completely disjointed.

The data set was recorded in different urban scenarios, mainly side roads and parking spaces. These scenarios contain approximately 1500 car and 5000 other objects. Table I lists the number of cell occurrences for each class in the data set.

Since there is no natural orientation of occupancy grids, the images are rotated randomly by up to 360° to increase the number and the variety of the training samples. Further the data were scaled by ± 5 percent, to improve the robustness with respect to the measurement error.

The value range of an occupancy grid is $[0, 1]$. The value 0 indicate *free space* while 1 ensures *occupancy* of the cell. For object classification only occupied cells are used, therefore values below 0.5 are set to 0 and the other cells $c^{[n]}$ are rescaled.

$$c^{[n]} = \begin{cases} 0 & \text{when } p(m^{[n]} = 1) < 0.5 \\ (2p(m^{[n]} = 1) - 1) & \text{otherwise} \end{cases} \quad (5)$$

III. CLASSIFICATION

For classification a pixel based learning framework[16] is used. The network was trained on the GPU on Linux work-

stations. In the next two sections the network architecture and the training method are explained.

A. Neural Network Structure

A fully convolutional neural network as introduced by Long et al. [10] is used as the classifier. The largest used CNN consists of 5 convolutional layers as depicted in Fig. 3. A *rectified linear unit* is used for activation. Except for the last layer, the convolutional layers interchange with max-pooling layers where the pooling size is set to 2x2 and the stride to 2. The last layer is fully connected with output neurons each of which represents one class. The last activation is a soft-max function. Other architectures with 2, 3 and 4 layers were examined. These networks use only the first n of the first 5 convolutional layers of the previous defined CNN. The last layer remains unchanged. Descending to deeper layer, the number of convolutional filters increase, whereas the size of the filters decreases.

When using fully convolutional layers, it is necessary to choose the input size of a training sample so that the image in the final layer of the network has a 1x1 dimension. The resulting input sizes of the network depends on the number of convolutional layer, for the 2, 3, 4, 5 convolutional layer architecture the input sizes are 22x22, 40x40, 60x60, 100x100. Hence the network takes into account different input areas for the prediction.

B. Training

TABLE II
TRAINING PARAMETERS OF THE CNN.

parameter	value
no. of convolutional layers	2, 3, 4, 5
learning rate	0.001, ..., 0.05
no. of filters of first layer (k)	16, 32, 64
weight decay	0, ..., 0.02
momentum	0.9
minibatch per epoch	20
samples per minibatch	5120
resulting size of ROI	22, 40, 60, 100

The network is trained in multiple epochs using a stochastic gradient descent algorithm. During a single epoch, a subset of images I_e from the image pool I is selected. For each iteration, a number of cells with their region of interest (ROI) are selected from I_e , where the size of the ROI equals the input size of the convolutional neural network (CNN). The choice is made randomly in such a manner that the classes occur evenly. After a certain number of stochastic gradient descent iterations a new epoch starts and a new subset of images is selected from the image pool.

When the classes subjected to training do not occur evenly in the input images, less strongly represented classes are not learned by the network. This is the case with the provided occupancy grids based on radar data where only roughly 2% of the image are labeled as cars and the rest as empty space, structures, vegetation or noise. As a counter measure the cells were balanced by class-specific selection during each iteration such that each class occurs evenly.

Table II summarizes all important training parameters.

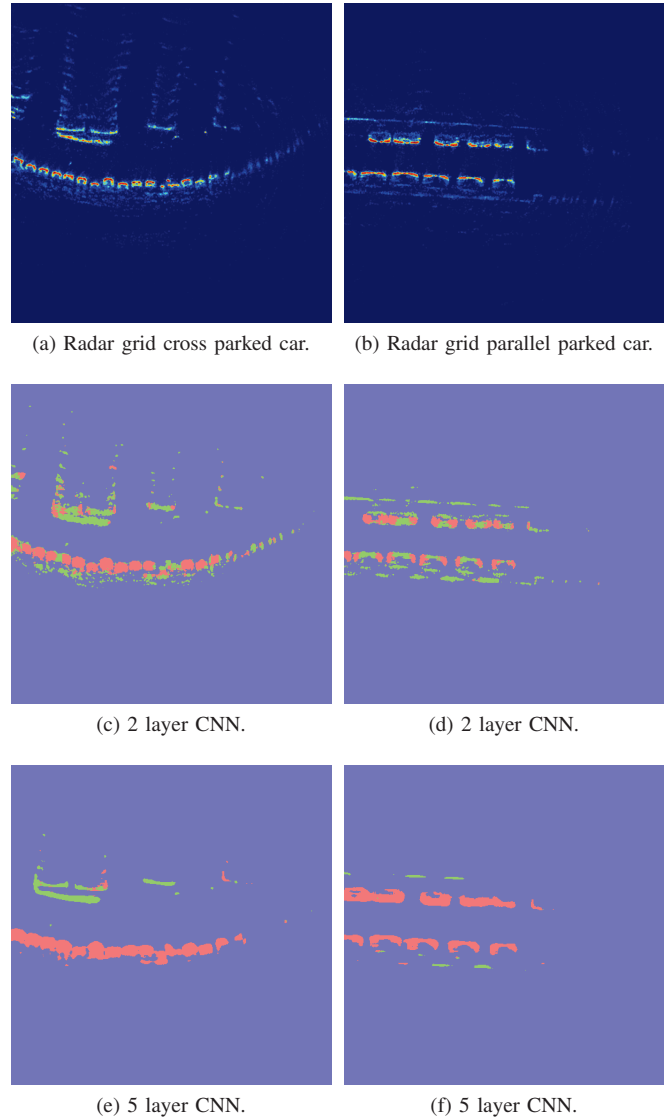


Fig. 5. The figure shows some example classification results for the 2 and the 5 layer CNN. On the left side cross parked and on the right side parallel cars can be seen. The red area are region where the classifier classified the cell as car and the green areas as non-cars. The 5 layer CNN shows better results, especially in the less characteristic parts of the cars. The number of filters are the number of filters of the first layer, the number of filter of the following layer is derived from there see Fig. 3.

IV. EVALUATION

The CNN is trained on the data set described in Section II-C. The training was stopped when the error rate on the validation set did not improve significantly any more. Next, after comparing the different designs, the best CNN design will be selected for a more detailed evaluation.

A. Comparison of Network Architecture

Fig. 4 shows the true positive and false positive rates of the different network designs, on pixel bases. The number of filters is given for the first layer. The numbers of the other filters are derived from that value. It stands out that the number of layers and the resulting larger ROI has the most significant

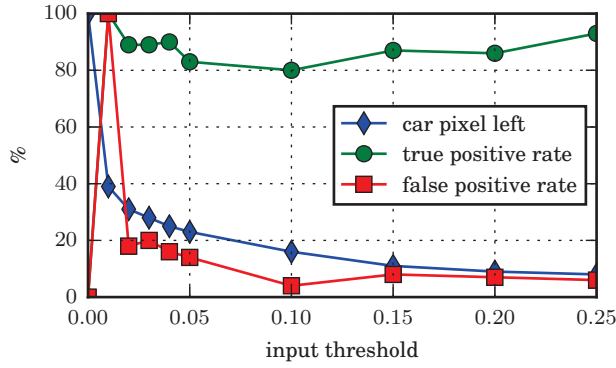


Fig. 6. Cell-wise results of the 5 layer CNN with respect to the occupancy threshold. The classifier was trained with 32 filters in the first layer.

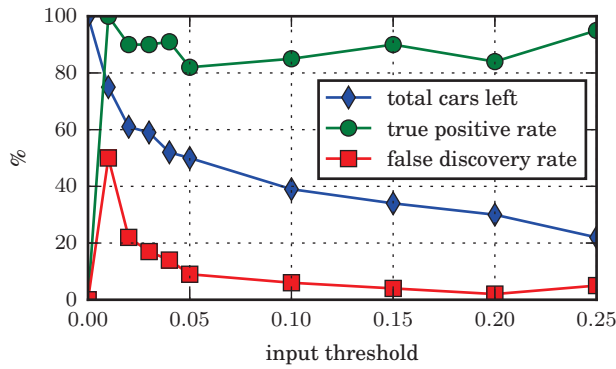


Fig. 7. Object-wise results of the 5 layer CNN with respect to the occupancy threshold. The classifier was trained with 32 filters in the first layer.

influence on the network. In Fig. 5 the results of the 2 layer and the 5 layer CNN on some parallel and some cross parked cars can be seen. This emphasizes that a larger input window lead to better results. The number of filters in the first layer, and thereby the number of filters in the subsequent layers, has only a minor effect on the performance. This reassures that the depth of the network and respective the size of the input window improves the classifier results and not the increasing number of parameters of the network.

B. Cell-wise Evaluation

The cell-wise results with regard to the occupancy threshold are plotted in Fig. 6. The number of car pixels left are depicted in blue, this gives the ratio between the considered car pixels and the car pixels when the occupancy threshold T_{occ} is set to 0. The rapidly decreasing values result from cars which are not fully built up in the grid and from the fact that only parts of the car can be seen by the sensor, while the label contains the full car. The classifier trained with a low occupancy threshold have poor classification results. Although not the whole real occupied area is considered, this is no disadvantage, e.g. for determining reliable landmarks.

C. Object-wise Evaluation

Depending on the use case of the classifier, a cell-wise evaluation may not be advantageous. In some cases the percentage of detected cars as whole objects is more informative. Therefore, the classifier is also evaluated based on the number of correctly classified vehicle objects. For an object-wise evaluation a criterion that decides which cells belong to a car and when the network should recognize a vehicle in the first place must be defined. This criterion depends on three parameters: First, the already introduced *occupancy threshold* T_{occ} that determines the minimum value in the occupancy grid to assign the label *car* instead of *not-labeled*. Secondly, a *car size threshold* T_{size} that decides how many cells have to be connected, with respect to an eight-connected neighborhood, in order to be recognized as a car. Thirdly, a threshold T_{cdp} that determines how many cells of a labeled car the classifier has to detect in order for it to be counted as found. This parameter is called the *car detection percentage*. For the following results the parameter set as follows: $T_{occ} = 0, \dots, 0.25$, $T_{size} = 100$ cells and $T_{cdp} = 0.7$.

Unlike the cell-wise evaluation, the computation of a false positive rate is not possible since it would require a number of true negative cars which is non-existent in this scenario. In order to still have a meaningful connection of true positives and false positives the false discovery rate[17] is used. It is defined as

$$\text{false discovery rate} = \frac{\text{false positives}}{\text{false positives} + \text{true positives}}. \quad (6)$$

In Fig. 7 the results of the object-wise classification are shown. Since the false discovery rate is based on the total number of found cars, no matter if true or false, a constant false discovery rate in combination with an increasing true positive rate means that the absolute number of false positives increases by roughly the same number. This means the number of false positives and true positives correlates with the object-wise evaluation as much as with the cell-wise evaluation.

V. CONCLUSION

This paper presents a cell-wise classification of radar grids and has shown that a semantic representation is possible with radar. Even on the limited training data classification with true positive rates of 80% and above were achieved. This is also true if the object-wise result is of interest. We have pointed out that the size of the ROI has a large influence on the classification results. False classification could be reduced by introducing an occupancy threshold, leaving out objects which are observed sparsely. The next logical step is to extend this classifier for multi-class classification by building up an appropriate data set. Here classes shown in [18], e.g. poles and curbstones should be considered. Also, estimation of the extent of an object would be desirable to determine the real shape of an object, e.g. for finding a proper parking spot.

VI. ACKNOWLEDGMENTS

The research leading to these results has received funding from the European Commission Seventh Framework Programme (FP/2007-2013) under the project AdaptIVe, grant agreement number 610428. Responsibility for the information and views set out in this study lies entirely with the authors. The authors would like to thank all partners within AdaptIVe for their cooperation and valuable contribution.

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